

Impedance Control Required:

1) DDR3_CK_P / DDR3_CK_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: 8 (BCU)

Reference: Plane above L7(GND3)

2) DDR3_CKE0, DDR3_RAS, DDR3_BA1, DDR3_CAS

Type: Single-ended

Target: $50\Omega \pm 10\%$

Layer: 8 (BCU)

Reference: Plane above L7(GND3)

3) DDR3_A1, DDR3_A5, DDR3_A8, DDR3_CS, DDR3_RESET

Type: Single-ended

Target: $50\Omega \pm 10\%$

Layer: 1 (FCU)

Reference: Plane below L2(GND)

4) DDR3_A0, DDR3_A2, DDR3_A3, DDR3_A4, DDR3_A7, DDR3_A14 DDR3_BA0,
DDR3_WE, DDR3_ODT0

Type: Single-ended

Target: $50\Omega \pm 10\%$

Layer: 4 (SIGNAL)

Reference: Plane above L3 (PWR) and plane below L5 (GND2)

5) DDR3_A6, DDR3_A9, DDR3_A10, DDR3_A11, DDR3_A12, DDR3_A13, DDR3_BA2

Type: Single-ended

Target: $50\Omega \pm 10\%$

Layer: 6 (SIGNAL2)

Reference: Plane above L5 (GND2) and plane below L7 (GND3)

6) DDR3_DQS0_P / DDR3_DQS0_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: 4 (SIGNAL)

Reference: Plane above L3 (PWR) and plane below L5 (GND2)

7) DDR3_D[0:4], DDR3_D6, DDR3_DM0

Type: Single-ended

Target: $50\Omega \pm 10\%$

Layer: 8 (BCU)

Reference: Plane above L7(GND3)

8) DDR3_D5,DDR3_D7

Type: Single-ended

Target: $50\Omega \pm 10\%$

Layer: 4 (SIGNAL)

Reference: Plane above L3 (PWR) and plane below L5 (GND2)

9) DDR3_DQS1_P / DDR3_DQS1_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: 4 (SIGNAL)

Reference: Plane above L3 (PWR) and plane below L5 (GND2)

10) DDR3_D[8:15], DDR3_DM1

Type: Single-ended

Target: $50\Omega \pm 10\%$

Layer: 1 (FCU)

Reference: Plane below L2(GND)

11) HDMI_TX_D0_P / HDMI_TX_D0_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L6 (SIGNAL2)

Reference: Plane above L5 (GND2) and plane below L7 (GND3)

12) HDMI_TX_D1_P / HDMI_TX_D1_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L8 (BCU)

Reference: Plane above L7(GND3)

13) HDMI_TX_D2_P / HDMI_TX_D2_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L6 (SIGNAL2)

Reference: Plane above L5 (GND2) and plane below L7 (GND3)

14) HDMI_TX_CLK_P / HDMI_TX_CLK_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L6 (SIGNAL2)

Reference: Plane above L5 (GND2) and plane below L7 (GND3)

15) HDMI_RX_D0_P / HDMI_RX_D0_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L1 (FCU)

Reference: Plane below L2(GND)

16) HDMI_RX_D1_P / HDMI_RX_D1_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L1 (FCU)

Reference: Plane below L2(GND)

17) HDMI_RX_D2_P / HDMI_RX_D2_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L1 (FCU)

Reference: Plane below L2(GND)

18) HDMI_RX_CLK_P / HDMI_RX_CLK_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L8 (BCU)

Reference: Plane above L7(GND3)

19) ETH_TD0_P / ETH_TD0_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L1 (FCU)

Reference: Plane below L2(GND)

20) ETH_TD1_P / ETH_TD1_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L1 (FCU)

Reference: Plane below L2(GND)

21) ETH_TD2_P / ETH_TD2_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L1 (FCU)

Reference: Plane below L2(GND)

22) ETH_TD3_P / ETH_TD3_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L1 (FCU)

Reference: Plane below L2(GND)

23) USBHOST_P / USBHOST_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L1 (FCU)

Reference: Plane below L2(GND)

24) USBHOST_P / USBHOST_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L8 (BCU)

Reference: Plane above L7(GND3)

24) USB_FTDI_D_P / USB_FTDI_D_N

Type: Differential

Target: $100\Omega \pm 10\%$

Layer: L1 (FCU)

Reference: Plane below L2(GND)

